Chapter 2

Trends in mobility
Credit card frauds in mobile and wireless Computing Era, Security Challenges posted by mobile devices

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Trends in Mobility

- In Generic , Mobility is moving the user with a device.
- In Mobile computing mobility has a broader meaning.
- ex - mobile , laptop, tab etc

1. Physical Mobility

2. Logical Moblity

1. Physical Mobility

1. User Mobility

User is able to move from one physical location to another location & use the same services.

2. Terminal (device) Mobility

Use to able to move from one device to another & use the same services.

2. Logical Mobility

1. Session Mobility

User session is able to move from one user agent environment to another

2. Service Mobility

Use to able to move from one service to another & service and transfer data.

Popular attacks against 3G mobile networks

1. Malware, Viruses & Warms
   - Skull Trojan
   - Cabir Warm
   - Mosquito Worm
   - Brador Worm
   - Lasco Worm

2. Dos
3. Overbilling Attack
4. Spoofed policy Development Process
5. Signaling level attacks,

Credit Card Frauds in Mobile & Wireless computing Era

- Working Principal of credit Cards

1. Plastic Money
2. Pre approved amount

Different between Credit Cards & Debit Cards

	Debit Card	Credit Card
Source of funds	Your bank account	A short-term loan from your bank.
Payment	Directly to the merchant as you're using your own money.	The bank pays the merchant, then you pay the bank back.
Interest	None	Interest is charged on the outstanding amount if it hasn't been paid by the due date.
Rewards	None	Enjoy cash back, points, miles, rewards, and more.
Fees	May have to pay annual fees or potential overdraft fees.	May have to pay for annual fees, late fees, cash advance fees, and more.
Builds credit score	No	Yes

Types of Credit card Frauds

1. Lost of stolen cards
2. Account takeover
3. clone
4. Never Reccived
5. Fraudulent Applications
6. Multiple Imprints
7. Collusive Merchants
8. Mail/Telephone order fraud

How to protect from Credit card Frauds

1. Don’t give your account details to anyone
2. Carry your creds separately from your wallet & carry only if you need for that outing.
3. During a transaction, keep your eye on your card.
4. Never sign on blank receipt.
5. Always take receipts from seller
6. Keep banking notification on.
7. Inspect POS machine before swapping

Security Challenges posted by mobile devices

1. Physical Security
2. Multiple User Logging
3. Secure data storege
4. Mobile Browsing
5. Application Isolation
6. System Updates
7. Mobile device coding issues
8. Bluetooth Attacks
9. Malware on the rise
10. Serious Threats in new Feature

How to ensure our Mobile Security & Privacy:

1. Alwasys password protected your phone.
2. Always read the terms & privacy policy
3. Only downlod safe apps.
4. Turn off the Bluetooth
5. Encrypt your phone
6. Set up remote locate/wipe
7. Backup your data
8. Don’t jailbreak or root your phone
9. Update OS
10. Downlode anti-malware
11. Use public WiFi with caution
12. Use a VPN (Virtual Private network)